Stage 6 — Check Your Understanding

Q1. Counting parameters

AICc = 2k – 2■ + 2k(k+1)/(n–k–1), with n = 33.

Caputo (k=5, ■=–161.49):

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AICc = 10 + 322.98 + 60/27 = 335.2022.
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Heterogeneous recovery (k=6, ■=–163.16):

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AICc = 12 + 326.32 + 84/26 = 341.5508.
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Difference = 341.5508 − **335.2022 = 6.3485** ≈ **6.4.** Caputo is preferred by the nominal AICc calculation. But the heterogeneous-recovery fit drives c toward infinity (the numerical boundary), so that parameter is not reliably estimated as an interior degree of freedom. Standard AICc parameter-count penalties assume regular, identifiable interior optima. Thus k=6 and the resulting difference should be regarded as approximate, not conclusive evidence of biological memory.

Q2. Reading Figure 19

Figure 19 compares the winning AICc model in ten synthetic datasets per mechanism at negative-binomial dispersions k_nb=10 (more noise) and k_nb=100 (less noise).

Truth	k_nb = 10	k_nb = 100
T2: declining transmission	True model 6/10; Caputo 3/10	True model 10/10
T5: Caputo α=0.6	Caputo 6/10; heterog. 3/10	Caputo within 6–9/10*
T7: heterog. recovery	Caputo 4/10 (40%); heterog. 1/10 (10%)	Caputo 1/10 (10%); heterog. 5/10 (50%)

*The paper states 6–9 of 10 Caputo wins across T4/T5, without specifying T5's exact low-noise count in its prose. The T7 counts are explicitly stated.

At dispersion around 2, counts are noisier than either simulation setting; mechanism identification is therefore especially unreliable. Proposition 11 establishes an exact equivalence between AHM fractional infectious-period survival and a memoryless mixture of exponential recovery rates. The same observed pattern need not imply unique memory. The mpox fitted dispersion is much higher (about 250–310), so this particular high-noise comparison chiefly concerns the other outbreaks.

Q3. Rolling origin and leakage

At each origin o = 8, 10, 12, …, each model fits only observations through o and forecasts up to four later weeks. Check 8 verifies that changing later observations does not change the forecast from a fixed prepared input. Nevertheless, future information enters in three ways:

1. Final revised data: historical counts may contain later corrections unavailable at the time of the forecast.

2. Whole-series cleaning: Ebola isotonic adjustment and interpolation use the complete reported series; earlier prepared weekly counts can depend on subsequent observations.

3. Retrospective take-off: the start week is determined using the full series, including the peak and preceding quiet runs.

The forecasts are therefore **pseudo-real-time**: model fits are truncated at the origin, but preparation is retrospective. The distinction matters more for Ebola because of irregular cumulative reports, revisions, isotonic cleaning, interpolation and delayed take-off; measles is already recorded as weekly cases and all three measles series have take-off at week zero, although revisions remain possible.

Q4. Good forecasts, wrong interpretation

For Short Creek, Table 14 gives mean CRPS ratio Caputo/ordinary SEIR = 0.919, corresponding to (1−0.919)×100 = **8.1% lower mean CRPS**. Yet α is at the lower grid edge in 18 of 21 forecast origins, compared with $\alpha = 0.86$ on the full series (Figure 22).

The additional order parameter can flexibly bend the predicted epidemic trajectory, especially the decline, improving forecasts without measuring a stable biological mechanism. Its movement with the starting week indicates that it acts as a shape parameter rather than an invariant transmission property.

Public-health recommendation. A team may use Caputo SEIR as a supplementary forecasting model when rolling-origin performance is better than ordinary and competing memoryless models. It should check the stability of α across origins, compare proper scoring rules and use uncertainty-calibrated forecasts. It should **not** claim that $\alpha<1$ proves biological memory, or that a boundary-hitting α measures a fixed transmission property. Plug-in forecast intervals are too narrow: nominal 90% intervals cover only 54–65% of observations and nominal 50% intervals cover 28–32%. Parameter uncertainty, for example via bootstrap or likelihood profiling, must be incorporated before relying on interval forecasts for decisions.